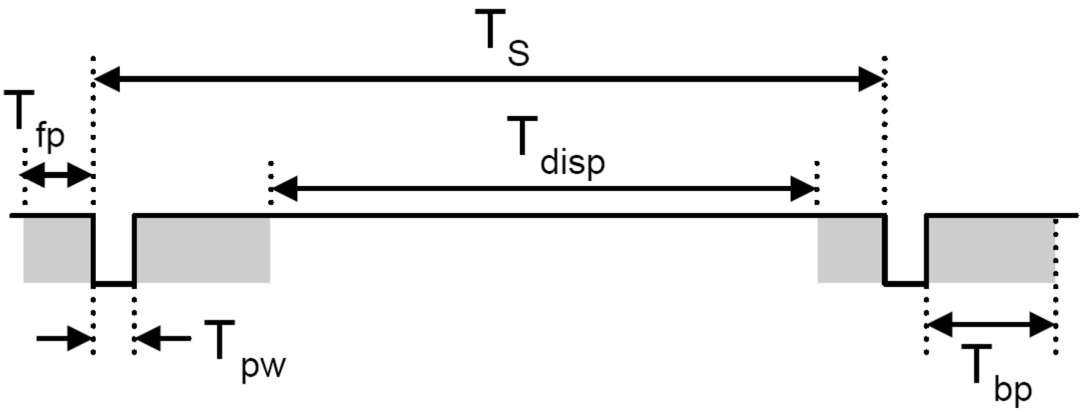
Lab 4 Hints

ECE 525.642 FPGA Design using VHDL

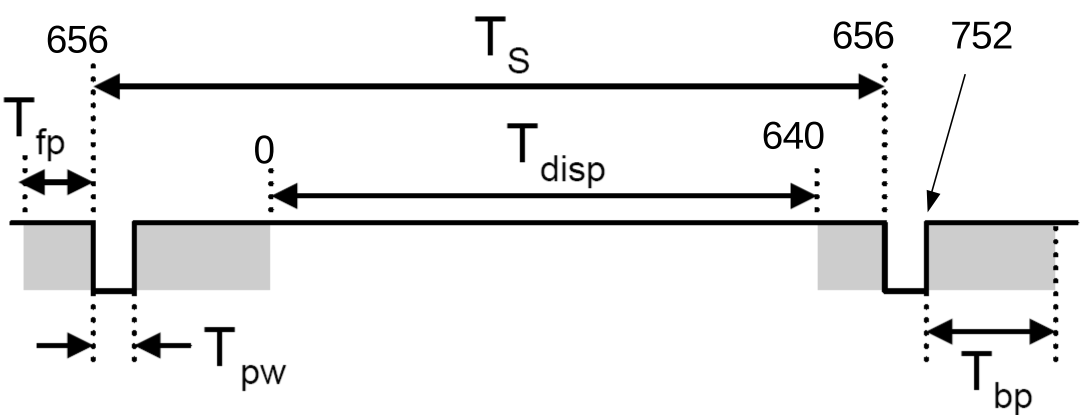
VGA Timing in VHDL

- Generate a 25 MHz **enable pulse** using a 100 MHz clock. Assume the pulse is high for one period of the 100 MHz clock and low otherwise. Name the pulse *en25.*
- Generate the skeleton/frame for a sequential process driven by the rising edge of the 100 MHz clock, enabled using the 25 MHz **pulse** *en25*. Assume the sequential process is reset by an asynchronous signal.
- In that process, generate a 10-bit counter named *horizontal_counter* that counts from 0 to 799 and resets to zero, upon which it starts counting again from 0 to 799.
- In that same process, generate a 10-bit counter named *vertical_counter* that counts that counts from 0 to 520. The vertical counter increments every time the horizontal counter resets. The vertical counter counts to 520, after which it resets and starts counting again from 0 to 520.
- Write a CSA to assign ‘0’ to a signal named *Hsync* when *horizontal_counter* is between [656,752) otherwise assign ‘1’.
- Write a CSA to assign ‘0’ to a signal named *Vsync* when *vertical_counter* is between [490,492) otherwise assign ‘1’.
- Write a combinational process to assign ‘1’ to a signal named *vgaRedT* when *horizontal_counter* is between [0, 640) and vertical counter is between [0, 480), otherwise assign ‘0’.
- Write a CSA to assign ‘0’ to signals *vgaGreenT* and *vgaBlueT.*
- Write a CSA to assign *vgaRedT* to a 4-bit output signal *vgaRed(3 downto 0).*
- Write a CSA to assign *vgaGreenT* to a 4-bit output *vgaGreen(3 downto 0).*
- Write a CSA to assign *vgaBlueT* to a 4-bit output *vgaBlue(3 downto 0).*

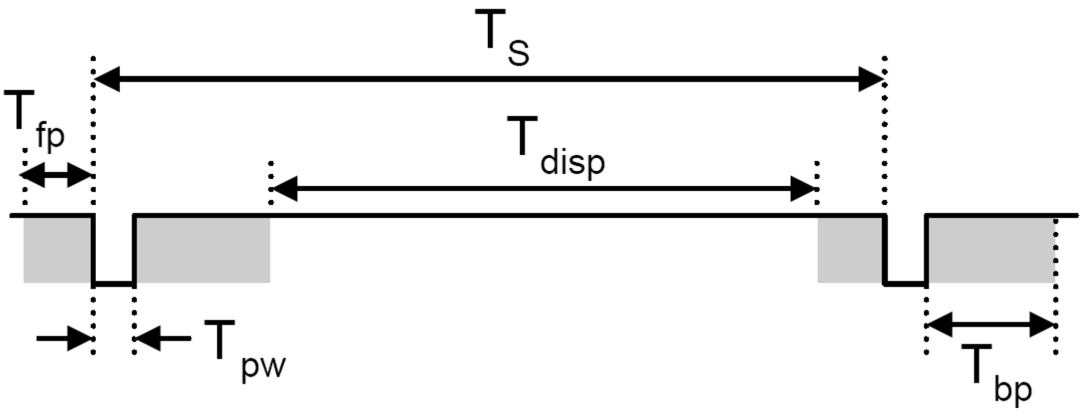
Horizontal Sync



Horizontal Counter values of interest



Vertical Sync



Vertical Counter values of interest